



Explaining frequency of verb morphology in early L2 speech

Roger Hawkins^{*}, Gabriela Casillas

*Department of Language and Linguistics, University of Essex, Wivenhoe Park,
Colchester, Essex CO4 3SQ, United Kingdom*

Received 18 July 2006; received in revised form 4 January 2007; accepted 4 January 2007

Available online 29 June 2007

Abstract

In speech, early L2 learners of English have been observed to supply forms of copula *be* more frequently than auxiliary *be*, and both more frequently than affixal regular past *-ed* and 3rd person singular present tense *-s* in contexts where morphological marking is required for native speakers. Early learners also use a construction not found in input: *be + bare V* (e.g. *I'm read*), allow constructions involving *be* to have a range of meanings not found in target English, and rarely overgeneralise *-ed* and *-s* to inappropriate contexts. The present study considers the kind of mental representation that L2 learners must have that would lead to the observed performance. A 'nativist' account is proposed. It is argued that the mental grammars of early L2 learners are organised in the same way as the grammars of native speakers, this following necessarily from the architecture of the language faculty. They differ minimally in the nature of their Vocabulary entries for verb morphology. This difference correlates with an early under-determination of syntactic representations where 'uninterpretable' syntactic features are absent from syntactic expressions. Evidence from a sentence completion task conducted with low proficiency speakers whose L1s are Chinese and Spanish is used to test this claim.

© 2007 Elsevier B.V. All rights reserved.

Keywords: Second language; Verb morphology; Input frequency; Distributed morphology; Contextual Complexity Hypothesis

^{*} Corresponding author.

E-mail addresses: roghawk@essex.ac.uk (R. Hawkins), Gabriela_casillas@yahoo.com (G. Casillas).

Table 1

Tense/agreement morphology in obligatory contexts (based on Tables 1 and 2 in Ionin and Wexler, 2002:106–107)

	Cop be	Aux be	Regular past	3p –s
Suppliance	329/431 (76%)	300/479 (63%)	73/174 (42%)	67/321 (21%)
Bare v	69/431 (16%)	158/479 (33%)	101/174 (58%)	250/321 (78%)
Tense/agreement mismatch	33/431 (8%)	21/479 (4%)	0/174 (0%)	4/321 (1%)

1. Observations requiring explanation

There have been a number of observations about the distribution of English verb morphology in the speech of early L2 learners. Three of these have been known since the early days of empirical research into second language acquisition:

- Forms of *be* (*I'm hungry*) are supplied more frequently than affixal forms (*She walks, walked*).
- Bare verb stems alternate with inflected forms in contexts where native speakers require inflected verbs (*Yesterday she walk home/Yesterday she walked home*).
- When inflected forms are used, there is little mismatch in S-V agreement (few cases of **I walks home*), or use of inflected past tense verbs in non-past contexts (*Now I played*).

A study by Ionin and Wexler (2002) of the speech of 20 L1 Russian child learners of English (age range 3,9–13,10) with varying lengths of immersion in English illustrates these 3 observations (Table 1). The instances recorded in Table 1 are obligatory contexts of use for native speakers. Observe that frequency of suppliance here does not simply divide between free forms and affixal forms, but distinguishes copula *be* from auxiliary *be*, and regular past from 3rd person singular present tense *–s*:

most frequent copula *be* < aux *be* < *–ed* < 3p *–s* **least frequent**

A number of other studies have also found dissociation in frequency of suppliance between copula and auxiliary *be*, and between affixal *–ed* and *–s*: for example, Dulay and Burt's (1974) study of child L1 Spanish and L1 Cantonese learners of English, Andersen's (1978) study of adult L1 Spanish learners of English, and a recent study by Paradis (2006) of 24 child learners of English from a variety of L1 backgrounds.

Two more recent observations should be added to this list. Ionin and Wexler (2002) note that their subjects produced a construction they would not have encountered in input: *be* + bare V,¹ e.g.:

I'm read
I'm buy beanie baby

Other studies have also observed this use of *be* + bare V by early L2 learners of English (Yang and Huang, 2004; García Mayo et al., 2005). This construction is used with a range of meanings other than the progressive/future readings of *be*+V-*ing* of native English (*I'm reading at the*

¹ For expository purposes, we treat the verb phrase as if it contained the single head V.

Table 2

Range of meanings of *be + bare V* (Table 4 in Ionin and Wexler, 2002:112)

	Prog	Generic	Stative	Past	Future	Ambiguous	Non-Prog/Fut
Tokens	32	33	12	21	5	5	71
%	30	31	11	19	5	5	66

e.g. They *are* help people when people in trouble (generic). He *is* run away. I stayed there (past).

Table 3

Range of meanings of *be+V-ing*, *be + bare V* and *bare V-ing*: AY: Ionin corpus – CHILDES (MacWhinney, 2000)

	Prog	Generic	Stative	Past	Future	Ambiguous	Total
Be + V-ing	11	6	0	6	2	3	28
Be + V	1	10	3	7	2	3	26
V-ing	1	5	0	2	0	0	8

moment, *She's leaving tomorrow*). In Table 2 the frequencies of these uses by subjects in the Ionin and Wexler study are displayed.

Since Ionin and Wexler's group covers children whose exposure to English ranged from 2 months to 2 years, we looked at this phenomenon of *be + bare V* in the speech of one of the informants with the least exposure to English: AY (from the Ionin corpus in the CHILDES database (MacWhinney, 2000)). AY was sampled twice by Tania Ionin, at age 10;1 after she had been in the USA for 2 months, and again at 10;4 (i.e. after 5 months of immersion). Data from both samples have been aggregated. It turns out that AY not only produces *be + bare V*, but also *be + V-ing* and *V-ing* on its own with a range of meanings not found in the input, as the frequencies in Table 3 illustrate. These constructions are used more with non-progressive/non-future meanings by AY than with the progressive/future meanings of the presumed input model *be + V-ing* (Table 4).

The distributional freedom of *be*, *V-ing* and *bare V* forms contrasts strikingly with the constrained use by AY of past tense forms (regular and irregular) and 3rd person singular present tense *-s*: 21/21 verbs inflected for past tense occur in intended past tense contexts, and 12/12 verbs inflected for *-s* appear in intended 3rd person singular present tense contexts. Whereas free forms are overused in a range of non-target contexts, affixal forms are only used by AY in obligatory contexts.

A further observation, also due to Ionin and Wexler (2002:108), is that the frequency of suppliance of verb morphology by these Russian speakers is unlikely to be the result of L1 influence. Russian has affixal inflections in all tenses, but lacks a copula in the present tense, and only has an equivalent of auxiliary *be* in the compound future. Frequency of forms in the L1

Table 4

Progressive/future and other meanings of *(be+V-ing)/be+V*

	Progressive/Future	Other
Be + V-ing	13/28	15/28
Be + v	3/26	23/26
V-ing	1/8	7/8
Total	17/62 (27%)	45/62 (73%)

appears to be quite different from frequency of forms in the L2. To this could be added the observation that a similar distribution of forms in speech has been found in speakers of typologically different L1s – Cantonese, Spanish and others – suggesting that generally L1 influence is an unlikely factor in the representation of English verb morphology by early L2 learners.

García Mayo et al. (2005) have shown that Basque/Spanish bilingual children learning English in the classroom produce *is + bare V* in oral narratives, ranging in frequency from 9% to 1% of all utterances involving verbs (and depending on the age at which subjects started learning English). They suggest that this construction (as well as constructions involving redundant pronouns such as *The mother he is on the tree*) is a realisation of the rich verbal inflections of Basque/Spanish. In other words, this is a form of transfer from the L1 where *is* is a ‘placeholder’ for tense/agreement affixes in the two L1s (2005:447). Although this explanation for the use of *be + bare V* by early L2 learners could presumably be extended to Russian speakers, there is evidence that such constructions also appear in the early English of L2 speakers of L1s that lack affixal tense/agreement morphology on the verb. Yang and Huang (2004) find a similar use of *be + bare V* in the written narratives of Cantonese classroom learners of English in Hong Kong. Their least proficient group ($n = 270$) produced *be + V* in 23% (45/191) of contexts involving unaccusative verbs, and 9% (164/1821) of other verbal contexts. Since Cantonese does not have affixal tense/agreement morphology on verbs, the ‘placeholder’ account does not appear to extend to this case.

The question these observations raise is: what is the nature of grammatical representation in early-stage L2 learners that would give rise to such patterns of performance?

2. Possible explanations based on salience and frequency of forms in input

Some emergentist accounts claim that properties of the input encountered by language learners play a major role in determining the kinds of knowledge they establish (and that underlies their own production) (Ellis, 2002). The postulation of pre-existing innate linguistic knowledge is unnecessary. Learners arrive initially at grammars on the basis of identifying recurrent patterns of form-meaning associations in what they hear or read. For example, Goldschneider and DeKeyser (2001:36) offer an account of the L2 acquisition of grammatical morphology that makes ‘no appeal to any innate blueprints or specific syntactic models . . . to explain order of acquisition’. On the basis of a ‘meta-analysis’ of 12 existing studies, they claim that properties of the input alone ‘account for a very large portion of the total variance in the accuracy scores for grammatical functors’ (2001:35) without the need to invoke pre-existing linguistic knowledge. The properties in question are perceptual salience, semantic complexity, morpho-phonological regularity, syntactic category and frequency in the input, which all ‘constitute aspects of salience in a broad sense of the word’ (2001:35). The problem with ‘salience’, however, is that it is an aspect of input that might lead learners initially to notice (or not notice) particular forms. The phenomena we are dealing with have all been ‘noticed’ by the learners studied since they are all represented in their speech. The question is why, having identified and stored forms of *be*, *-ing*, past tense and 3rd person singular present tense *-s*, they are produced differentially. Notions of semantic complexity, morpho-phonological regularity and so on, are not sufficient on their own to account for why one known form is supplied less frequently than another.

One way in which Goldschneider and DeKeyser’s proposal might be interpreted is that the infrequent forms of regular past *-ed* and 3rd person singular present tense *-s* in the speech of

early L2 learners have in fact *not* been noticed as independent morphemes by early L2 learners. Their presence as part of stored forms is just ‘noise’ (Larsen-Freeman, 2002:280). However, if such forms were just noise, random probabilistic use across contexts might be expected, but this is not the case. Tense and agreement affixes are largely used appropriately, although infrequently, as we have seen.

Frequency of forms alone in input is often cited as the *prime* determinant of frequency in the speech of early L2 learners. Paradis (2006) outlines one such account, based on the ‘network model’ of Bybee (2001). Bybee proposes that single- and multi-morphemic words are stored fully inflected in the lexicon and are associatively connected to other lexical items which share phonology and semantic features. The frequency of tokens in the input (and also frequency in the speaker’s own output) determines the lexical strength of a stored form. Tokens that are more frequent in input/output will cause higher levels of activation of the stored form than tokens that are less frequent. Type frequency in input/output – that is, the frequency with which the same property is realised by different tokens belonging to the type (e.g. *–s* in *writes*, *walks*, *hits*, . . .) – increases the likelihood that a ‘schema’ (= a rule) for the property in question will be formed.

Paradis collected data from 15 child L2 learners of English in Canada after 9 months, 21 months and 34 months of exposure to English, and compared the frequency and the accuracy with which plural *–s* with count nouns (e.g. *book-s*) and 3rd person singular present tense *–s* (e.g. *write-s*) occurred in their speech. At each sampling, the children were producing more instances of plural noun contexts than 3rd person singular present tense contexts, and within those contexts were producing a higher proportion of plural *–s* than 3p *–s*, both token and type. To consider whether this pattern might be the effect of frequency in the input, Paradis examined the distribution of plural *–s* and 3p *–s* in the British National Corpus (a large collection of written and transcribed oral texts). The pattern of token and type frequency in the corpus, and the pattern of token and type frequency in the speech of the L2 learners is remarkably similar.

On the face of it, this looks like striking evidence that what early L2 learners of English know about properties like plural *–s* and 3p *–s* is a direct function of how often they encounter these forms in input. However, consider the detail of what is involved. The input to learners from native speakers contains, presumably, *categorical* marking of plural in plural contexts (i.e. 100% of regular plural count singular nouns are of the form N-*s*) and *categorical* marking of 3p singular in 3p singular present tense contexts (i.e. 100% of regular verbs are of the form V-*s*). Learners are not exposed to input containing forms like **Two book*, **She write* (except, perhaps, in ‘slip of the tongue’ contexts). Frequency of forms like *books* and *writes* in native speech and writing is relative to other tokens in the sample. For early L2 learners, however, the frequency of suppliance that we are concerned with is the relative frequency of N-*s* to bare nouns in obligatory N-plural contexts, of V-*s* to bare verbs in obligatory V-3p singular contexts. To illustrate, suppose, hypothetically, that the token frequency of plural *–s* is 10%, and the token frequency of 3p singular *–s* is 5%; the difference between what this means in a sample of native speech and a sample of early L2 speech is as follows:

<i>Native sample</i>	<i>Early L2 samples</i>
N- <i>s</i> / All tokens = 10%	N- <i>s</i> / tokens of N-plural = 10%
V- <i>s</i> / All tokens = 5%	V- <i>s</i> / tokens of V-3p-sing = 5%

The claim must then be that L2 learners convert frequency of N-plural and V-3p-sing relative to the whole set of lexical items they encounter into ‘probable suppliance’ of N-*s* in contexts where N-plural is required and V-*s* in contexts where V-3p-sing is required. How this kind of

conversion would work is not obvious. One possible construal is that relative frequency of categorically marked forms to total number of lexical items encountered in input is converted into the strength of activation of the memory trace for the item. The activation level of plural *-s* is higher than the activation level of 3rd person singular present tense *-s*, and both levels of activation are weak relative to the activation levels of bare N and bare V. However, activation of this type would be blind to distributional constraints on N-*s* and V-*s*. As was seen in the results in Table 1, both 3p singular *-s* and regular past *-ed* (another form probably with relatively low frequency in input) are not used randomly, but restricted to contexts where the subject is a 3rd person singular noun or the verb is in the context of a T[+past]. An ‘input frequency’ account, without further qualification, predicts the probability of V-*s* and V-*ed* occurring in any context. The account would need to be supplemented by constraints such as:

- (1) a. /s/ [5% probability] ↔ [V, -past, 3p, +sing]+____
 b. /d/ [20% probability] ↔ [V, +past]+____

But, of course, specifications like (1) presuppose that learners have existing knowledge of properties like tense, number and person that are used for analysing input prior to any encounter with input. We conclude that proposals that input frequency predicts output frequency in the speech of early L2 speakers are making hidden assumptions that learners have pre-existing knowledge of the syntactic properties involved.

3. An explanation that assumes innate linguistic knowledge

There have been a number of hypotheses that offer potential explanations for the observations under consideration here, and that assume that L2 mental grammars develop within a space defined by innate linguistic knowledge: Zobl and Liceras (1994), Schwartz and Sprouse (1996), Haznedar and Schwartz (1997), Prévost and White (2000), Ionin and Wexler (2002), among others. For reasons of brevity, these proposals will not be discussed here. Rather, we outline another hypothesis: that the observed differences in frequency of suppliance of verb morphology in the speech of early L2 speakers is an effect of the way that phonological exponents are initially stored in the ‘Vocabulary’ component of the grammar, and not an effect of the frequency or salience of forms in input. This account also assumes that L2 learners use innate linguistic knowledge to deduce grammatical representations from linguistic experience.

Our proposal involves two elements: (1) assumptions about the language-faculty-determined organisation of grammatical representation common to humans; (2) differences between early L2 speakers of English and native speakers in the way they implement that organisation.

4. Organisation of the grammar

We assume that the architecture of the language faculty determines a uniform organisation of grammatical representation in humans, including L2 learners. The relevant aspects of this organisation for our purposes are:

- (a) UG provides a finite set of *interpretable* syntactic features (Chomsky, 2000) that guide learners in their categorisation of linguistic experience. The availability of this set dramatically limits the range of possible hypotheses about grammatical representation that a language learner might entertain. For example, in the case of subject-verb agreement, a

learner is faced with a large range of possible factors that might be relevant for agreement: animacy, human-ness, agency, whether the subject is derived or not (e.g. *learner* versus *pupil*), social status of the subject (e.g. *professor* versus *student*), size of the subject (e.g. *elephant* versus *mouse*), whether the verb is in a matrix or embedded clause, whether the subject ends in a consonant or a vowel, whether the subject is a possessive or not (*my book* versus *the book*), sex of the subject, grammatical Case of the subject, respect due to the subject, grammatical gender of the subject, person, number, and a potentially vast list of alternatives. It turns out that where languages realise agreement properties on verbs, the only features that are potentially relevant are person, number, grammatical gender and ‘respect’, i.e. honorific markers (Corbett, 2006) and even here there is considerable controversy about whether honorific markers are really cases of agreement (Corbett, 2006:137). Pre-existing knowledge of the interpretable features that are potentially relevant to computing subject-verb agreement in the form of the UG feature inventory greatly reduces the hypotheses that a learner has to entertain, allowing rapid convergence on the target grammar.

- (b) UG offers a further subset of *uninterpretable* syntactic features. These determine agreement and movement in mature native grammars (Chomsky, 2001). Uninterpretable features are valued by cognate interpretable features (Pesetsky and Torrego, 2001) in a local, c-command domain. Although uninterpretable features are invisible to the semantic component (hence their ‘uninterpretability’) they may have morpho-phonological reflexes in the forms of the exponents that realise them. For example, for native speakers of English the agreeing form of the verb in *Her brother writes* is a reflex of uninterpretable person and number features of T that have been valued by *her brother* and merged with the verb. The terminal node [V, WRITE, *u*:3p, *u*:+sing], where [*u*:3p] and [*u*:+sing] are the valued uninterpretable person and number features of T respectively, is then the input to the ‘Vocabulary’ where phonological exponents are stored (see (c) and (d) below).
- (c) Grammars instantiate some version of the *separation hypothesis* (Beard, 1995) where phonological exponents are separated from the expressions that are the output of syntactic operations, are stored in a ‘Vocabulary’ component, and are inserted into syntactic terminal nodes only after all syntactic operations have applied, as in distributed morphology (Halle and Marantz, 1993; Harley and Noyer, 1999; Embick and Marantz, 2005; Embick and Noyer, 2007).
- (d) Phonological exponents have entries in the Vocabulary which specify their *contexts of insertion*. For example, native speakers of English might have entries like:

- (2) /bʊk/ ↔ [N, BOOK]
- /(t)z/ ↔ [T, BE, –past, +sing, 3p]
- /m/ ↔ [T, BE, –past, +sing, 1p]
- /s/ ↔ /[V, –past, +sing, 3p]+ ____
- /d/ ↔ /[V, +past] + ____

Insertion occurs through feature-matching between the exponent and the terminal node. Matching does not require full feature identity, hence Vocabulary entries may be under-specified by comparison with syntactic terminal nodes. For example, the Vocabulary entry /d/ in (2) would be inserted into a terminal V node specified not only for [tense], but also for [person] and [number] (features required for the insertion of –s).

- (e) Some statements of the context of insertion of a Vocabulary item are context-free (e.g. /bʊk/, /vz/), others are *context-sensitive* (e.g. for affixes like /s/, /d/). The fact that entries in the Vocabulary allow context-sensitive statements for insertion will play an important role in our proposal for how grammatical organisation is implemented in early L2 grammars.

5. Differences between early L2 grammars and the grammars of native speakers of English

A parsimonious assumption is that the organisation of interlanguage grammars (ILGs) is the same as the organisation of the grammars of mature native speakers, and that assumption is made here. Our proposal is that there is a specific difference, however, in the representation of entries for phonological exponents in the Vocabulary. For native speakers, Vocabulary items are specified in terms of bundles of features of the point of insertion (i.e. the terminal node), with limited context-sensitivity, as in (2).

Non-native speakers at an early stage of acquisition, by contrast, have Vocabulary entries for exponents realising dependencies (such as subject-verb agreement, agreement between T and the thematic verb, agreement between auxiliary *be* and *V-ing*) that are context-sensitive. They are statements about the terminal nodes with which an exponent co-occurs, rather than statements about the features of a terminal node into which the form is inserted. To illustrate, the kind of contrast we have in mind between a native entry for 3rd person singular present tense *-s* and an early L2 entry is shown in (3):

- (3) Native speaker Vocabulary entry:
 /s/ ↔ [V, -past, +sing, 3p]+ ____
 L2 speaker Vocabulary entry:
 /s/ ↔ /[V]+ ____ / [T, -past] ____ / [N, +sing, 3person] ____

The L2 speaker entry is understood as ‘insert /s/ in the context of a verb which is in the context of a non-past T, itself in the context of a 3rd person, singular N’.

Underlying this proposal is the assumption that L2 learners initially have access to the interpretable syntactic features of the UG inventory, and assign these to phonological strings extracted from the input stream; so a noun like *brother*, once identified, is assigned the features [N, 3p] as part of its Vocabulary entry. By contrast, L2 learners do not initially have access to uninterpretable features. So phonological properties that realise dependencies, like 3rd person *-s* and forms of *be*, are not initially associated with syntactic features. Yet they are identified as recurrent and stable phonological strings in input. Their representation in the Vocabulary is in the form of context-sensitive rules specifying the nodes with which they cooccur.

Hypothetically, then, a set of early L2 learner Vocabulary entries for English verb morphology might look like those in (4):

- (4)
- | | | | |
|-----|------------------|---|---|
| a. | /s/ | ↔ | /[V]+ ____ / [T, -past] ____ / [N, +sing, 3person] ____ |
| b. | /d/ | ↔ | /[V]+ ____ / [T, +past] ____ |
| c. | /ɪŋ/ | ↔ | /[V]+ ____ |
| d. | {ran, rəʊt, ...} | ↔ | [V, RUN, WRITE ...] / [T, +past] ____ |
| e. | /m/ | ↔ | [T, BE] / [N, +sing, 1p] ____ |
| e’. | /z/ | ↔ | [T, BE] / [N, +sing, 3p] ____ |
| f. | {wo:k, raʊt ...} | ↔ | [V, WALK, WRITE] |

6. Frequency of forms in speech is a function of early Vocabulary entries: the Contextual Complexity Hypothesis

We now propose that the different frequencies of forms in early L2 speech is an effect of the storage of Vocabulary items with context-sensitive contexts of insertion. The idea is that context-sensitive entries are costly to access if more than one terminal node is involved, leading to the following hypothesis.

Contextual Complexity Hypothesis (CCH)

The probability with which a Vocabulary item is retrieved during the derivation of a syntactic expression is a function of the number of sister terminal nodes required to specify the context in which it is inserted. The more sister nodes required to specify the context, the greater the probability that the entry will not be retrieved.

Given the entries in (4), the CCH predicts that bare V forms are highly likely to be retrieved, since entries for Vs require no statement about sister terminal nodes, while 3rd person singular present tense /s/ is highly unlikely to be retrieved because its entry involves three sister terminal nodes. The likelihood of regular past tense /d/ being retrieved is greater than 3rd person /s/ (because the statement for its insertion involves only two sister terminal nodes) but lesser than forms of *be*, *-ing* or irregular past tense forms. This broadly corresponds to the frequency patterns for suppliance of these forms by early L2 learners described in section 1, although it does not explain dissociation in the frequency of forms of copula and auxiliary *be*. See below for discussion.

This proposal also gives some account of the observations in section 2, which are summarised here:

- (a) Production of the *be* + bare V construction for which there is no model in input (e.g. *I'm read*).
- (b) Allowing *be* + bare V, *be* + V-ing and V-ing to have a range of meanings other than progressive/future (the meanings assigned to *be* + V-ing by native speakers).
- (c) Lack of overgeneralisation of /s/ and /d/.

The production of *be* + bare V follows as a possibility from the entries proposed in (4) because the statement of the contexts of insertion of /m/ and /z/ (allomorphs of *be*) refer only to T and the properties of the left-adjacent node in the terminal string. In a syntactic terminal string of the form N-T-V, the form of the Vocabulary entries would allow both insertion of forms of *be* and either bare V or V-ing. Forms of *be* are inconsistent, however, with /s/ and /d/ (**She's walks*, **She's walked*). This is because learners appear to specify contexts of insertion for /s/ and /d/ early on in terms of a T that is itself specified for [+past], but not for [BE].

Since none of /m/, /z/ or /ɪŋ/ has any specification beyond their immediate contexts of insertion, they can be used with a range of meanings. By contrast, the requirement that /s/ occur in the context of a T[–past] and a 3rd person singular N to the left, and that /d/ occur in the context of a T[+past] precludes their use with other meanings.

If this account is along the right lines, the early grammars of L2 speakers of English are minimally different from those of mature native speakers. They are organised in the same way, but differ in the nature of the Vocabulary entries. The contexts of insertion of phonological exponents are specified in terms of co-occurring syntactic terminal nodes for the L2 speakers, but

in terms of the features of the node into which the exponent is inserted for the native speakers. The probability that L2 speakers will omit forms in speech increases as a function of the complexity of the context-sensitive statement in its entry.

7. Later restructuring of early Vocabulary entries

It was observed above that dissociation in the frequency of production of forms of copula and auxiliary *be* does not automatically fall out from the proposed analysis. Yet such dissociation has been found in a number of studies (see, for example, the results from Ionin and Wexler (2002) presented in Table 1). Interestingly, dissociation does not appear in the productions of L2 speakers with the earliest grammars, but emerges later. In AY's sample (recall that AY had had 5 months of immersion) the frequency of forms of *be* in obligatory copula contexts is 24/30 (80%) and in auxiliary contexts 12/14 (86%). It appears that the dissociation arises as a result of subsequent restructuring of the grammar. This raises the question of how early context-sensitive Vocabulary entries restructure towards the target, and we consider this here.

Taking a hypothetical example, an L2 learner needs to move from an entry for a 1st person singular form /m/ of the kind in (5a) to a native entry of the kind in (5b):

- (5) a. /m/ ↔ [T, BE]/[N, +sing, 1p] ____
 ↓
 b. /m/ ↔ [T, BE, -past, ~~unumber~~: +sing, ~~uperson~~: 1]

(5b) is different from (5a) in two ways. First, it reflects the fact that the entry is now specified for the interpretable tense feature [+/-past] of T, whereas initially the entry for /m/ was not (as shown by the range of tense interpretations possible in AY's early productions: generic, past and future). Second it reflects the fact that T has been assigned the uninterpretable syntactic features [*unumber*] and [*uperson*], and that the syntax has assigned values to these features and deleted them from the expression that goes for semantic interpretation.

It is the first of these changes, we speculate, – the identification of a new interpretable syntactic feature – that gives rise to a dissociation in frequency of copula and auxiliary forms of *be*. Learners come to recognise from input that /m/, /s/ realise a progressive interpretation when they co-occur with a V-ing form. The entry for /m/ changes from (5a) to the two entries in (6):

- (6) a. /m/ ↔ [T, +prog]/[N, +sing, 1p] ____ V-ing
 b. /m/ ↔ [T, -prog]/[N, +sing, 1p] ____

(6a) specifies that when the clause is interpreted as progressive, /m/ must co-occur with a V-ing form. Since the entry for this /m/ now involves a context specified in terms of two terminal nodes, the likelihood of this /m/ being retrieved is lesser than the likelihood of non-progressive /m/ being retrieved, which only has one node specified in the statement of its context of insertion.

An interesting question is how Vocabulary entries change to allow uninterpretable features, like [*unumber*] and [*uperson*], to be specified. Speculating again, here frequency of forms in the input (or output that the learners produce themselves) plays a crucial role. Because context-sensitive Vocabulary entries are costly to access (given the Contextual Complexity Hypothesis) they may only be retrieved infrequently in a learner's grammar in the early stages of acquisition. However, as more tokens of entries are encountered over time, the activation level of these entries will increase. That is, although the Contextual Complexity Hypothesis predicts that the more

terminal nodes involved in the specification of an entry's context of insertion, the greater the probability it will not be retrieved, as the number of tokens encountered increases over time, so will the level of activation of the Vocabulary entry. This captures the well-known observation that the frequency with which L2 speakers supply verb morphology in speech gradually increases as they become more proficient. Input/output frequency, then, plays a role in raising the activation levels of context-sensitive Vocabulary entries. Our speculation is that some critical level of activation of a context-sensitive Vocabulary entry is reached where it triggers the selection of an uninterpretable syntactic feature from the UG inventory. For example, repeated activation of the entries in (6) triggers the selection of [*u*number] and [*u*person] which are then assigned to T. T is one of the categories that provides input to syntactic computations. Therefore, frequency information from the Vocabulary component must be available to items that are stored in the lexicon and which form the lexical arrays selected as input to syntactic computations.

This has interesting consequences in theories of ultimate attainment in a second language which assume that the mental grammars of older learners may be representationally deficient. In previous work, Hawkins (Hawkins and Liszka, 2003; Hawkins and Franceschina, 2003; Hawkins and Hattori, 2006) has argued that uninterpretable syntactic features may be subject to a critical period (following ideas developed by Smith and Tsimpli, 1995; Tsimpli, 2003). Where an uninterpretable feature has not been selected for grammatical representation during primary language acquisition in early childhood, that feature disappears from the feature inventory. In the context of the present proposal this view suggests that whereas all L2 learners of English will have early grammars involving context-sensitive Vocabulary items, they may diverge when entries reach the critical level of activation for restructuring. The grammars of speakers whose L1s contain the relevant uninterpretable feature will restructure, while the grammars of speakers of L1s that do not will continue with context-sensitive Vocabulary entries. This would have the potential performance effect of L2 speakers showing persistent optionality where their L1 grammars lack the relevant feature, but speakers of L1s that have the relevant feature achieving categorical supplience. Such differences in performance in advanced L2 speakers are widely reported in the literature.

8. Testing the proposal

Our proposal for the early Vocabulary entries of L2 speakers is that dependencies are formulated as context-sensitive statements about the position a form is inserted into in the string of syntactic terminal nodes. If this idea is correct, it should make the following prediction:

Prediction: Where a Vocabulary entry is specified in terms of a string of adjacent terminal nodes, if that string is disrupted by some extraneous node(s) in the syntactic expression that is input to Vocabulary insertion, retrieval of the form will be affected.

For example, the likelihood of 3rd person *-s* being retrieved in a context like (7b) will be different compared with (7a):

- (7) a. My brother owns a house
- b. The brother *of my best friend* owns a house

In (7b) a PP complement intervenes between the N determining agreement and T. By contrast, the likelihood of 3rd person *-s* being supplied in (7c) will be the same as in

(7a), even though a complex subject is involved. In this case the N determining agreement and T are adjacent:

- c. My best friend's brother owns a house

Disruption in (7b) might take two forms. Firstly, if a learner's Vocabulary entry for /s/ is determined by strict linear adjacency (i.e. the entry refers to N_2 in the sequence N_1 – of – PRN – A – N_2 – T – V), insertion will be sensitive to N_2 , and not the *real* subject of the sentence, which is N_1 . Thus if N_2 is singular, suppliance of –s on the verb should be the same as when single N subjects are singular, whatever the number of the *real* subject. Secondly, if the learner's Vocabulary entry is sensitive to the headedness of the subject (i.e. the entry refers to N_1 and not N_2) then the expectation is a greater tendency not to supply –s. The reason for this is that the terminal nodes that have to be computed to retrieve the Vocabulary entry for /s/ are no longer adjacent because of the intervening PP, increasing the computational burden and making the retrieval of the entry for /s/ less likely.

For native speakers, the Vocabulary entry for /s/ is specified in terms of features of V itself, as illustrated in (2). The disruption of the string of terminal nodes by some extraneous node(s) should have no effect on the activation of /s/.

8.1. Materials

To investigate this prediction, an experiment was conducted which was a modified version of the sentence completion task used by Bock and Miller (1991). Participants were shown a lexical verb (e.g. *own*) or an adjective (e.g. *blond*) on a computer screen for 2 seconds. This was replaced by an intended subject of a sentence (a 'preamble') which remained on the screen for 4 seconds. The preamble was either a simple DP (*my brother*) or a complex DP where the N determining agreement was followed by a PP complement (*The brother of my best friend*) or preceded by a genitive modifier (*My best friend's brother*). When the preamble disappeared from the screen, the participant's task was to utter a complete sentence aloud, beginning with the preamble and using either the stimulus verb or adjective.

The role of the simple DP preamble was to provide a baseline measure of frequency of suppliance of copula *be* (if the stimulus was an adjective) or –s (if the stimulus was a lexical verb). The experimental items were those involving a complex DP preamble: (i) where a PP complement disrupted adjacency between the N determining agreement and T; (ii) where a preceding genitive DP did not disrupt the adjacency. The distribution of items in the test is shown in Table 5.

The head nouns of simple DPs were presented in either their singular (S) or their plural (P) form. For the complex subject preambles, modifier noun phrases were presented in both their singular and their plural forms, resulting in four number conditions: singular–singular (SS), singular–plural (SP), plural–plural, (PP) and plural–singular (PS). (8) illustrates sample preambles.

Table 5
Details of the sentence completion task

Preambles	Predicates
128 Simple DPs (<i>The guest</i>)	128 lexical verbs (<i>own</i>)
56 DP of DP (<i>The guest of my music teacher</i>)	128 adjectives (<i>blond</i>)
56 DP's DP (<i>My music tutor's guest</i>)	
36 fillers (<i>My brother and my friend</i>)	

- (8)
- | | |
|----|-------------------------------------|
| SS | The brother of my best friend |
| SP | The brother of my best friends |
| PP | The assistants of the math teachers |
| PS | The assistants of the math teacher |

Head nouns and verbs were on average separated by the same number of syllables in complex noun phrase preambles containing a prepositional phrase: 8 or 9 syllables. All head nouns included in the preambles were animate, but inanimate subject modifiers were introduced in half of the preambles. Collective nouns such as *group* or *committee* were not included, given that subject-verb agreement with this type of noun varies among native speakers. Stimulus verbs were selected on the basis of their inherent aspectual class—only stative verbs and psych-verbs were included.² The inclusion of this variable therefore reduces any effect of inherent lexical aspect on the production of 3rd person singular present tense marking.

8.2. Hypotheses

The hypotheses were the following:

H1. Early L2 learners of English will supply copula /(*ʌ*)z/ more frequently than 3rd person /s/ because they have Vocabulary entries like (4), and retrieval of exponents is subject to the Contextual Complexity Hypothesis; more terminal nodes need to be computed to activate /s/ than /(*ʌ*)z/. Native speaker controls will show no difference because they have Vocabulary entries specified in terms of features of the point of insertion.

H2. Separation of the N determining number and person agreement from T will lead either to a decrease in suppliance of /(*ʌ*)z/ and /s/ or to (mis-)agreement with the closest N by early L2 learners, but not by native speakers.

H3. Complex DP subjects *per se* will not affect the probability of insertion: DP's DP will not affect suppliance of /(*ʌ*)z/ and /s/.

8.3. Additional task

To determine whether participants were assigning the appropriate structural analysis to complex DPs (i.e. were correctly identifying the head N, the potential controller of agreement marking on the verb), they were given a supplementary 20-item comprehension test consisting of items of the following form:

- (9)
- | | |
|----|--|
| a. | Tom bought the friend of his sister a book.
Who got a present?
a) his sister's friend b) Tom c) his sister d) Tom's friend |
| b. | The tree in the garden was blown away in the storm.
What got damaged?
a) the garden b) the tree c) the storm |

² The need for this restriction was apparent after the results of a pilot study showed that for low intermediate speakers of L2 English the inherent lexical aspect of the verbs influenced their production of present tense marking, with stative verbs being more likely to be marked than dynamic verbs.

Table 6

Background details of the participants in the experiment

L1	N	QPT score range	QPT score mean	Age range	Age mean
Chinese	10	30–39	34.3	21–30	24.2
Spanish	10	31–39	35.4	22–35	27.9
English	10	–	–	18–35	26.2

If participants were assigning the appropriate structure to the complex object DP in (9a) they should choose answer (a), and in (9b), where there is a complex subject DP, they should choose answer (b). Results from this task provide supplementary evidence bearing on whether subjects are likely to make SV agreement between the closest N and the verb or between the head of the subject DP and the verb.

8.4. Participants

The non-native speaker participants were not complete beginners. It was doubtful whether complete beginners would be able to handle the demands of the task. They were 20 lower intermediate proficiency speakers of English (as determined by the Oxford *Quick Placement Test*, 2001), 10 native speakers of Chinese and 10 native speakers of Spanish, together with a control group of native speakers (Table 6). Since the hypothesis is that the early grammars of all L2 learners of English have Vocabulary entries where dependencies are specified in terms of co-occurring terminal nodes, the fact that the participants are adults who have typically had both formal instruction and immersion in English, and therefore contrast with the child learners discussed in section 1, should not have an impact on the results. The reason why L1 speakers of Chinese and Spanish were compared was that subject-verb agreement is realised in Spanish but not in Chinese. If properties of L1 verb morphology are influential in the early acquisition of English (contrary to expectations) this should show up in the performance of the two groups.

8.5. Results

The percentages of suppliance of /*(t)z*/ (with adjective stimuli) and /*s*/ (with lexical verb stimuli) on the sentence completion task are presented in Table 7.

Table 7

Mean % of suppliance of /*(t)z*/ and /*s*/

	Chi		Span		Eng	
	/i(t)z/	/s/	/i(t)z/	/s/	/i(t)z/	/s/
(a) The guest	100	80	100	81	100	100
(b) The guests	0	0	0	0	0	0
(c) My music tutor's guest	100	79	99	82	100	100
(d) The guest of my music tutor	99	60	100	61	100	100
(e) The guest of my music tutors	78	47	80	49	100	100
(f) The assistants of the math teacher	22	40	18	35	0	0
(g) The assistants of the math teachers	0	1	0	0	0	0

Table 8
Mean % scores on the sentence comprehension task across L1 groups

L1 group	n	Minimum score	Maximum score	Mean%	S.D.
Chinese	10	17	20	18.8	1.13
Spanish	10	17	20	18.7	1.06
English	10	17	20	19.4	1.07

The results show that for the native speakers there is no effect of the form of the subject on suppliance of /(*ʌ*)z/ or /s/, consistent with Vocabulary entries that are activated directly by a single syntactic terminal node.³

For both groups of L2 speakers, results are strikingly similar, suggesting again that the L1 is unlikely to be influential in determining the knowledge that gives rise to these patterns of response:

- (i) /(*ʌ*)z/ is supplied more than /s/ with simple subjects, consistent with H1. There is no overgeneralisation of /(*ʌ*)z/ or /s/ when the subject is plural.
- (ii) there is no decrease in the suppliance of /(*ʌ*)z/ or /s/ when there is a complex subject with a preceding genitive DP, consistent with H3.
- (iii) Suppliance of /s/ with a complex subject is disrupted when there is an intervening PP, consistent with H2. However, suppliance of /(*ʌ*)z/ is only disrupted where the PP contains a plural N, not when both Ns are singular. This is only partially consistent with H2.

Does disruption take the form of mis-agreement with the closest N, or omission resulting from the non-adjacency of the terminal nodes required for activation of the Vocabulary entries? In the case of /(*ʌ*)z/ it looks like mis-agreement with the closest N is involved because when *is* is not supplied, *are* is, rather than omission of *is*. When *-s* is not supplied, a bare V is. In the case of rows (e) and (f) in Table 7, this looks like it might also be the effect of mis-agreement with the closest N, which differs in number from the head N. However, mis-agreement with the closest N cannot explain the decrease in suppliance shown in row (d), where both Ns are singular.

Results from the supplementary comprehension task (Table 8) show that these informants have knowledge of the headedness of complex subjects and objects in English. A one-way ANOVA showed no difference between groups ($F_{2, 27} = 1.21$, $p = .32$).

Together the results suggest that where constituents disrupt adjacency of strings of terminal nodes specified in the entry of a Vocabulary entry, this leads either to retrieval of a form whose specification for the context of insertion matches an intervening (and inappropriate) node or failure to retrieve the entry.

³ Previous studies with native speakers by Bock and Miller (1991), Vigliocco and Nicol (1998) found some 'agreement attraction errors', e.g. *The helicopter for the flights are safe*, but in only 5% (B and M) and 6% (V and N) of cases, and mainly in the condition: DP_[singular]-P-DP_[plural]. The L2 subjects in the present study show some sensitivity to DP_[singular]-P-DP_[plural] contexts, but their responses are qualitatively different. Observe that the native controls in this study make no agreement attraction errors.

9. Discussion

The article began by outlining a number of observations about the production of English verb morphology by early L2 learners, and asking what kind of grammatical representation underlies such behaviour. The key observations are the following:

- In the speech of early-stage L2 learners of English, the frequencies of suppliance of copula *be*, auxiliary *be*, regular past *-ed*, and 3rd person singular present tense *-s* are dissociated.
- There is little overgeneralisation of *-ed* and *-s*.
- Early learners use a construction not found in input: *be + bare V*.
- early learners overgeneralise *be + V-ing*, *be + bare V* and *V-ing* to a range of meanings not associated with *be + V-ing* in the input they receive.
- transfer from the L1 is unlikely to be the source of the observations.

Our proposed account hypothesises that early learners have L2 grammars that are organised in the same way as those of native speakers, and this follows from the architecture of an innate language faculty: UG. However, learners differ from native speakers of English in the nature of their Vocabulary component, where phonological exponents of syntactic terminal nodes are stored with specifications of their contexts of insertion. Early L2 entries identify terminal nodes that co-occur with the node into which the item is inserted, whereas native speaker entries are activated by features of the node itself. Associated with a constraint inherent to the insertion operation – the Contextual Complexity Hypothesis – which proposes that the probability that a Vocabulary item will be retrieved decreases the more terminal nodes are required in its specification, it was suggested that the observations listed above broadly follow. Dissociation in the frequency of production of copula *be* and auxiliary *be* is a consequence of restructuring of Vocabulary entries to incorporate reference to an interpretable [+/-Progressive] feature. We attempted to provide further support for the account in an experimental sentence completion task with lower intermediate proficiency L2 speakers of English. This showed that the presence of an extraneous constituent between a subject N and T disrupted the suppliance of /s/ and to a lesser extent the suppliance of /ʌz/ for the non-native speakers but not for the native speakers. Non-native speakers tended either to make mis-agreements with the closest N, or omitted /s/ to a greater extent when the intervening constituent was present. This follows if their Vocabulary entries are as proposed.

The view underlying this approach is that there are two stages that L2 learners go through in converting linguistic experience into underlying representations.

- (a) Identification and storage of phonological (and/or orthographic) strings in the Vocabulary;
- (b) Establishment of connections between items in the Vocabulary and entries in the lexicon (where the lexicon is the store of items accessed by syntactic computations).

This is highly speculative at this stage, and requires further investigation. But there are two reasons for thinking the idea worth pursuing. The first is that there is evidence from L1 development that children may be aware of dependencies between phonological exponents and their sisters before they have fully determined the morpho-syntactic functions they realise (Stager and Werker, 1997; Johnson et al., 2005). If this is the case, it may also be the case for L2 learners. The second is that an initial conservatism in specifying the contexts of insertion for

Vocabulary items may have learnability advantages. Representing dependencies like subject-verb agreement *syntactically* requires invoking uninterpretable syntactic features that are valued by cognate interpretable syntactic features. Uninterpretable features are part of the set that enter syntactic derivations, and give rise to automatic and fully general agreement. However, in the early stages of experience with a new language, L2 learners do not yet know whether the apparent cases of agreement they are encountering are general or idiosyncratic. If learners invoke a syntactic uninterpretable feature immediately, and the property turns out not to be a general case of agreement, retreat from overgeneralisation becomes problematic. Constructing Vocabulary entries that are statements of superficial co-occurrence with other categories bearing interpretable syntactic features might be a low-risk learning device until sufficient evidence is accumulated to determine the generalisability of agreement.⁴

Acknowledgements

We would like to thank the audience at GASLA 2006 (Banff, Canada), William O'Grady and two reviewers for comments on an earlier version of this article. Its limitations are entirely our own responsibility.

References

- Andersen, R., 1978. An implicational model for second language research. *Language Learning* 28, 221–282.
- Beard, R., 1995. *Lexeme-morpheme Base Morphology*. SUNY Press, Albany, NY.
- Bock, J.K., Miller, C.A., 1991. Broken agreement. *Cognitive Psychology* 23, 45–93.
- Bybee, J., 2001. *Phonology and Language Use*. Cambridge University Press, New York.
- Chomsky, N., 2000. Minimalist inquiries. In: Martin, R., Michaels, D., Uriagereka, J. (Eds.), *Step by Step: Essays on Minimalist Syntax in Honor of Howard Lasnik*. MIT Press, Cambridge, MA, pp. 89–156.
- Chomsky, N., 2001. Derivation by phase. In: Kenstowicz, M. (Ed.), *Ken Hale: A Life in Language*. MIT Press, Cambridge, MA, pp. 1–52.
- Corbett, G., 2006. *Agreement*. Cambridge University Press, Cambridge.
- Dulay, H., Burt, M., 1974. Natural sequences in child second language acquisition. *Language Learning* 24, 37–53.
- Ellis, N.C., 2002. Frequency effects in language processing: a review with implications for theories of implicit and explicit language acquisition. *Studies in Second Language Acquisition* 24, 143–188.
- Embick, D., Marantz, A., 2005. Cognitive neuroscience and the English past tense: comments on the paper by Ullman et al. *Brain and Language* 93, 243–247.
- Embick, D., Noyer, R., 2007. Distributed morphology and the syntax/morphology interface. In: Ramchand, G., Reiss, C. (Eds.), *The Oxford handbook of linguistic interfaces*. Oxford University Press, Oxford.
- García Mayo, M.P., Lázaro Ibarrola, A., Licerias, J., 2005. Placeholders in the English interlanguage of bilingual (Basque/Spanish) children. *Language Learning* 55, 445–489.
- Goldschneider, J.-M., DeKeyser, R.M., 2001. Explaining the natural order of L2 morpheme acquisition in English: a meta-analysis of multiple determinants. *Language Learning* 51, 1–50.
- Halle, M., Marantz, A., 1993. Distributed morphology and the pieces of inflection. In: Hale, K., Keyser, S. (Eds.), *The View from Building 20*. MIT Press, Cambridge, MA, pp. 53–109.
- Harley, H., Noyer, R., 1999. Distributed morphology. *Glott International* 4, 3–9.
- Hawkins, R., Franceschina, F., 2003. Explaining the acquisition and non-acquisition of determiner-noun gender concord in French and Spanish. In: Paradis, J., Prévost, P. (Eds.), *The Acquisition of French in Different Contexts*. John Benjamins, Amsterdam, pp. 175–205.

⁴ A reviewer points out that most of the observations outlined in sections 1 and 2 come from studies of child L2 learners, whereas the experiment reported in section 8 involves adults, and that therefore the presupposition must be that child and adult L2 acquisition have the same characteristics. That is indeed our assumption as far as the development of early Vocabulary entries is concerned. This does not preclude the possibility that child and adult L2 learners differ on other aspects of development.

- Hawkins, R., Hattori, H., 2006. Interpretation of English multiple wh-questions by Japanese speakers: a missing uninterpretable feature account. *Second Language Research* 269–301.
- Hawkins, R., Liszka, S., 2003. Locating the source of defective past tense marking in advanced L2 English speakers. In: van Hout, R., Hulk, A., Kuiken, F., Towell, R. (Eds.), *The Lexicon-Syntax Interface in Second Language Acquisition*. John Benjamins, Amsterdam, pp. 21–44.
- Haznedar, B., Schwartz, B., 1997. Are there optional infinitives in child L2 acquisition? In: Hughes, E., Hughes, M., Greenhill, A. (Eds.), *Proceedings of the 21st Annual Boston University Conference on Language Development*. Cascadia Press, Boston, pp. 257–268.
- Ionin, T., Wexler, K., 2002. Why is 'is' easier than '-s'? acquisition of tense/agreement morphology by child second language learners of English. *Second Language Research* 18, 95–136.
- Johnson, V.-E., de Villiers, J.-G., Seymour, H.-N., 2005. Agreement without understanding? The case of third person singular /s/. *First Language* 25, 317–330.
- Larsen-Freeman, D., 2002. Making sense of frequency. *Studies in Second Language Acquisition* 24, 275–285.
- MacWhinney, B., 2000. *The CHILDES Project: Tools for Analyzing Talk*, third ed. Lawrence Erlbaum, Mahwah, NJ.
- Paradis, J., 2006. Differentiating between SLI and child SLA: focus on functional categories. In: Paper Presented at the Generative Approaches to SLA (GASLA) Conference. University of Calgary, Banff, Canada.
- Pesetsky, D., Torrego, E., 2001. T-to-C movement: causes and consequences. In: Kenstowicz, M. (Ed.), *Ken Hale: A Life in Language*. MIT Press, Cambridge, MA, pp. 355–426.
- Prévost, P., White, L., 2000. Missing surface inflection hypothesis or impairment in second language acquisition? evidence from tense and agreement. *Second Language Research* 16, 103–133.
- Quick Placement Test, 2001. Oxford University Press, Oxford.
- Schwartz, B.D., Sprouse, R., 1996. L2 cognitive states and the full transfer/full access model. *Second Language Research* 12, 40–72.
- Smith, N., Tsimpli, I.-M., 1995. *The Mind of a Savant*. Blackwell, Oxford.
- Stager, C.L., Werker, J.F., 1997. Infants listen for more phonetic detail in speech perception than in word-learning tasks. *Nature* 388, 381–382.
- Tsimpli, I.-M., 2003. Features in language development. In: Paper Presented at the European Second Language Association (Eurosla) Conference. University of Edinburgh.
- Vigliocco, G., Nicol, J., 1998. Separating hierarchical relations and word order in language production. Is proximity concord syntactic or linear? *Cognition* 68, B13–B29.
- Yang, S., Huang, Y., 2004. A cross-sectional study of be + verb in the interlanguage of Chinese learners of English. In: Paper Presented at the First National Symposium on SLA, Guangzhou, China.
- Zobl, H., Liceras, J., 1994. Review article: functional categories and acquisition orders. *Language Learning* 44, 159–180.